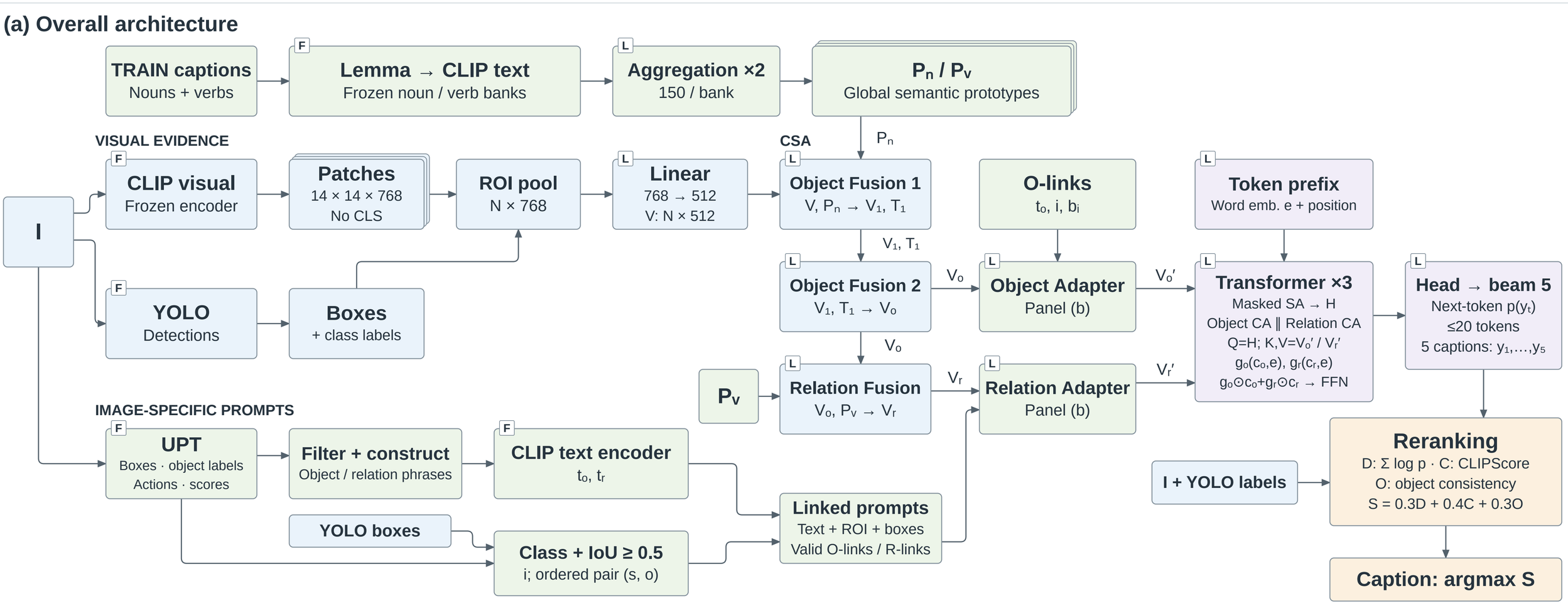

(a) Overall architecture
TRAIN captions
Nouns + verbs
F
Lemma → CLIP text
Frozen noun / verb banks
L
Aggregation ×2
150 / bank
Pn / Pv
Global semantic prototypes
VISUAL EVIDENCE
CSA
Pn
I
F
CLIP visual
Frozen encoder
Patches
14 × 14 × 768
No CLS
ROI pool
N × 768
L
Linear
768 → 512
V: N × 512
L
Object Fusion 1
V, Pn → V1, T1
O-links
to, i, bi
L
Token prefix
Word emb. e + position
V1, T1
F
YOLO
Detections
Boxes
+ class labels
L
Object Fusion 2
V1, T1 → Vo
Vo
L
Object Adapter
Panel (b)
Vo′
L
Transformer ×3
Masked SA → H
Object CA ‖ Relation CA
Q=H; K,V=Vo′ / Vr′
go(co,e), gr(cr,e)
go⊙co+gr⊙cr → FFN
L
Head → beam 5
Next-token p(yt)
≤20 tokens
5 captions: y1,…,y5
Vo
Pv
L
Relation Fusion
Vo, Pv → Vr
Vr
L
Relation Adapter
Panel (b)
Vr′
IMAGE-SPECIFIC PROMPTS
F
UPT
Boxes · object labels
Actions · scores
Filter + construct
Object / relation phrases
F
CLIP text encoder
to, tr
Reranking
D: Σ log p · C: CLIPScore
O: object consistency
S = 0.3D + 0.4C + 0.3O
I + YOLO labels
YOLO boxes
Class + IoU ≥ 0.5
i; ordered pair (s, o)
Linked prompts
Text + ROI + boxes
Valid O-links / R-links
Caption: argmax S
D, C: min–max over 5 beams · O: normalized · ties: earlier beam